Shui Whiteboard Generator

Direction update: the illustration engine, and a consolidated cost estimate

Prepared for: development team and shareholders
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Project: Shui WG (self-hosted whiteboard video generation, internal service)

One-line summary: the pipeline was rendering text-and-bullet-point videos with no real drawings. Phase 4 adds real image generation — two providers, built side by side — and keeps the whole project comfortably under Golpo's per-video cost even in the worst case.

Executive summary

Shui WG is the self-hosted alternative to Golpo AI, built to render Shui's whiteboard-style lesson videos at a fraction of Golpo's cost. The first four build phases (0–3) shipped a working pipeline — real narration, real rendering, a deployed API, multiple visual styles — but a review of actual output revealed the videos contained no real illustrations, only title cards, bullet lists, and timelines. That was a scoping gap in the original phase plan, not a bug: the scene planner was explicitly instructed not to reference images, because at the time nothing in the pipeline could produce one.

A new Phase 4 closes that gap. Per your direction, it builds **two** real image-generation providers behind one swappable interface — Recraft (vector illustration) and Flux Schnell (raster illustration) — rather than committing to one now. Both are wired into the same pipeline, both are cost-tracked per job, and a dedicated comparison script renders the same script through each so the final provider choice is made from real output and real numbers, not a guess.

Bottom line on cost: even in the least favorable case — no cache hits, the more expensive provider — a one-minute illustrated video costs an estimated \$0.10–\$0.27 through Shui WG, against \$2.00 through Golpo's API-only tier and roughly \$1.33 through Golpo's Growth plan. That is an 85–95% reduction versus Golpo, with real illustrations included, not the stripped-down text-only version that prompted this review.

What changed, and why

The problem

A sample render (a script about the Great Wall of China) was reviewed frame by frame. Every scene was typographic: a title card, a generic icon with bullet text, a text-only timeline, a plain bullet list. No drawings, figures, or diagrams of the actual subject matter appeared anywhere in the video.

The root cause

The scene-planning system prompt (`src/schema/planning.ts`) contained this instruction, verbatim: *“never use ‘documentReveal’ or ‘fullBleedGraphic’ — no images are available.”* The two Remotion components built to display full-frame illustrations already existed and worked correctly — they simply had never been given a real image to render, because Phases 0–3 never built an image-generation step. This was sound, honest engineering under the original scope; it just left the scope too narrow for a product called a “whiteboard video.”

The fix

Phase 4 adds a real image-generation step ahead of rendering, resolving a short text description (“imageConcept”) into an actual generated image before the video is composited — no changes needed to the two components that already knew how to display one.

Updated phase roadmap

The productization roadmap (previously Phase 4) was renumbered to Phase 5 to make room for the real illustration-engine phase. Nothing else in the sequence changed.

Phase	Delivers	Status
0 — Foundation	Repo scaffold, GCP setup, one hardcoded scene rendered locally and in Docker	Shipped
1 — Render pipeline	Scene schema, component library, real TTS timing sync, compositing, cost telemetry	Shipped
2 — API surface	Golpo-shaped REST API, async jobs, API-key auth, deployed on Cloud Run	Shipped
3 — Style library	Icon library, 3–4 visual styles, script-only scene planning via LLM	Shipped
4 — Illustration engine	**New.** Real image generation — Recraft + Flux, swappable, cached, cost-tracked	New — this update
5 — Productization roadmap	Public API on tri-nham.com, billing, docs site, rate limiting for strangers	Not started (design doc only)

Phase 5 is explicitly not a build phase yet — the trigger condition is sustained real production data (including real Phase 4 illustration costs) from Shui itself using Shui WG, not an estimate. Pricing, billing, and public launch decisions are deferred until that data exists.

Phase 4 technical summary (for the developer)

Full spec lives in `prompts/shui-wg/shui-wg-phase-04-illustration-engine.md` in the repo — this is a condensed version. Seven concrete pieces:

1. Scene schema addition

One new optional field on `SceneAction`: `imageConcept: string`. `documentReveal` / `fullBleedGraphic` now accept either a supplied `imageUrl` (human-authored) or an `imageConcept` (resolved by the new pipeline step below).

2. The provider interface

```
interface ImageProvider {
  readonly name: "recraft" | "flux";
  generate(concept: string, opts: {
    styleVariant: string;
    orientation: "vertical" | "horizontal";
  }): Promise<GeneratedImage>;
}
```

Two real implementations — `src/images/recraft.ts` and `src/images/flux.ts` — selected via `getImageProvider(name)`, same pattern as the existing `getTheme(styleVariant)`.

3. Content-addressable cache

Cache key = `sha256(provider + styleVariant + concept)`. Firestore doc + R2 object. A cache hit costs $0.00 and increments a hit counter. This is the main lever that keeps average cost below the worst-case numbers below as the content library grows and concepts repeat.

4. Pipeline hook point

New step in `resolveSceneDocument.ts`, immediately after scene parsing and before TTS — same position in the pipeline TTS itself already occupies. Resolves every `imageConcept` to a real `imageUrl`, cache-first, concurrency-capped.

5. Planner change

The system prompt in `planning.ts` no longer forbids the two image components. It now instructs the planner to use them sparingly — 1–3 illustrations per video, not every scene — and to describe what should be drawn, never to invent a URL.

6. Cost telemetry

JobCost gains imagesGenerated, imageCacheHits, imageGenerationCostUsd, and imageProvider fields. Every render logs a real, itemized cost — the same discipline already applied to TTS and render compute.

7. Comparison script

`scripts/render-illustration-comparison.ts` renders one fixed script through both providers and prints a real cost/quality table — the artifact that should actually decide which provider ships as default, mirroring the existing `render-style-comparison.ts` pattern.

Consolidated cost estimate

All figures below are derived from real, published per-unit rates already hardcoded in the codebase (Cloud Run compute rates, TTS character pricing, LLM token pricing) and from each image provider's published list price — not guesses. They will be superseded by real metered data once the Phase 4 comparison script has been run and Shui WG has real production volume.

Spend so far (local testing, Phases 0–3)

Roughly 14 local test renders have been produced across the existing test scripts (render-batch, render-style-comparison, render-script-only-batch). At the Phase 1–3 non-visual per-video rate below, estimated total spend to date is **approximately $0.20–$0.50**. This is an estimate reconstructed from render counts and published rates, not a pull from a real billing dashboard.

Per-video cost model, going forward

Approach	Cost / 1-min video	Notes
Shui WG — narration + render only (Phases 0–3, no illustrations)	$0.02 – $0.03	TTS + Cloud Run compute + optional scene-planning LLM call
Shui WG + Recraft illustrations (Phase 4, worst case)	$0.10 – $0.27	1–3 images/video @ ~$0.08 each, zero cache hits assumed
Shui WG + Flux illustrations (Phase 4, worst case)	$0.035 – $0.09	1–3 images/video @ ~$0.015–$0.02 each, zero cache hits assumed
Golpo — API-only tier	$2.00	$2.00/minute, $200 minimum, no dashboard access
Golpo — Growth plan	~$1.33	$200 / 2 months for 150 one-minute-video credits

Caching is expected to pull the illustration-inclusive numbers down further as the content library grows and concepts repeat across topics (e.g. common iconography reused across related civics or finance videos) — that effect is real but not yet measured, so it is deliberately left out of the worst-case figures above.

Savings versus Golpo

Comparison	Reduction
Shui WG (Recraft, worst case) vs. Golpo API-only tier	~87–95% cheaper
Shui WG (Flux, worst case) vs. Golpo API-only tier	~95–98% cheaper
Shui WG (Recraft, worst case) vs. Golpo Growth plan	~80–92% cheaper
Shui WG (Flux, worst case) vs. Golpo Growth plan	~93–97% cheaper

Even in the least favorable case — the more expensive provider, and no benefit yet from caching — Shui WG remains at least an order of magnitude cheaper than Golpo while producing videos with real illustrations, which the current Golpo-based output does not consistently guarantee either.

Open decisions and next steps

- **Provider choice is deliberately not made yet.** Run `render-illustration-comparison.ts` and review both outputs before picking Recraft, Flux, or a per-style-variant mix of both as the default.
- **Confirm image licensing** for commercial use under both providers' current terms before any public-facing use (internal-only use today is unaffected).
- **Track real cache-hit rate** once Phase 4 is live — this is the number that determines whether the worst-case cost figures above hold or trend down meaningfully.
- **Phase 5 (productization) stays paused** until Shui itself has generated a meaningful volume of real, illustrated production content through Shui WG — not before.

Full technical specifications: `prompts/shui-wg/shui-wg-phase-04-illustration-engine.md` *and* `prompts/shui-wg/shui-wg-README.md` *in the Shui WG repository prompt set.*
All cost figures are estimates derived from published provider rates and existing cost-telemetry code, pending real production measurement.